CONNEXT DDIL NETWORKING

Link Health From Protocol Statistics

Per-peer measurement, calibrated thresholds, and degradation inference

The DDIL link-health problem

A mesh link can be reachable while still failing to carry operational traffic reliably or within latency bounds.

Reachability

Peer presence determines whether a router or path is reachable.

Degradation

Loss, retransmission, delay, and queue pressure often appear before a peer becomes unreachable.

Decision

The router needs per-peer evidence to distinguish a degraded link from an unavailable one.

> Presence answers: “is the peer reachable?” Protocol statistics answer: “how is reliable delivery behaving?”

Measure at the Connext router



In-process telemetry

Connext exposes reliable-protocol status directly on writers and readers. No additional WAN monitoring protocol is required for these counters.

Stable collection

Poll cumulative totals at a fixed interval, calculate deltas locally, and key every record by peer and network.

Writer-side: repair demand and send pressure

`DataWriterProtocolStatus`

- received NACK count and bytes
- pulled samples and fragments for repair
- pushed samples and bytes for offered load
- oldest unacknowledged sequence and send-window size

Writer-cache and reader activity

- unacknowledged sample count and peak
- high-watermark and full-cache events
- current send-window capacity
- inactive matched readers

> Increasing NACKs and repair traffic identify demand for recovery. Persistent unacknowledged backlog and full-window events identify pressure on the sending side.

Reader-side: holes, rejection, and end-to-end loss

Gap recovery

NACKs sent identify missing sequence numbers detected at the receiver.

In-order delivery

Uncommitted samples show data held behind a missing earlier sample.

Resource limits

Out-of-range/rejected samples indicate local receiver pressure, not necessarily RF loss.

Failure signal

`SampleLostStatus::lost_by_writer` means the writer removed a sample before the reader recovered it. Keep this separate from local resource-limit losses and rejections.

Attribute every signal to one peer

Outbound view

Use matched-subscription protocol status for each remote reader rather than one aggregate writer total.

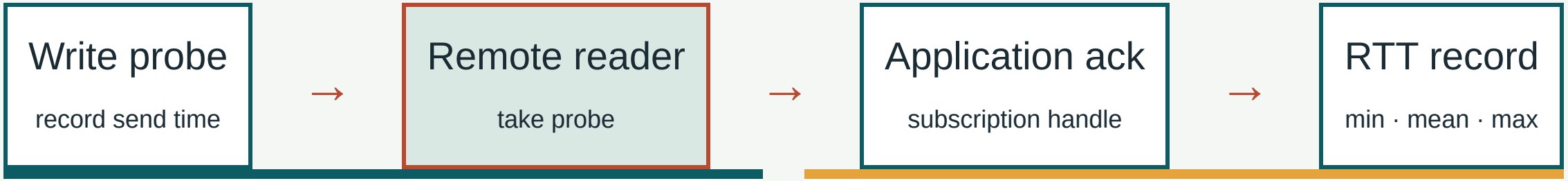
Inbound view

Use matched-publication protocol status for each remote writer rather than one aggregate reader total.

Endpoint lifecycle matters

A rematch, restart, or rediscovery creates a new endpoint association and can reset counters or cause durable replay. Mark that interval and reset its baseline; do not classify it as link recovery or degradation.

RTT requires an application-ack probe



Dedicated probe topic

A small, reliable, volatile probe topic isolates measurement QoS and retention from the presence authority and operational data routes.

Interpretation

Application-ack RTT includes scheduling, queueing, reader processing, acknowledgment timing, and the return path. It is a system round trip, not pure RF latency.

Normalize and combine the measurements

Loss and repair

NACK rate and repair ratio: NACKs/pushed samples, pulled/pushed samples, fragment repair for large data.

Congestion and queueing

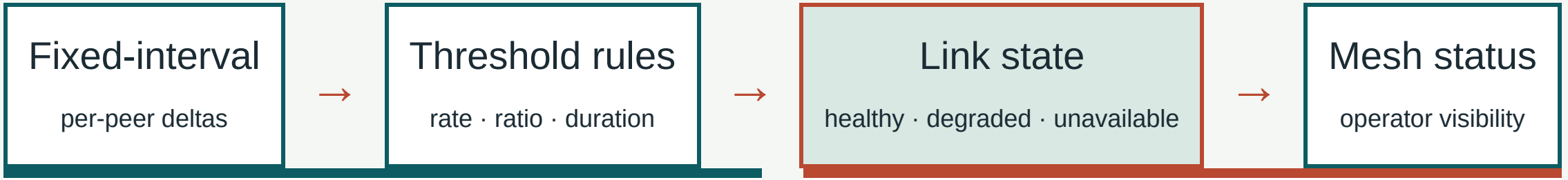
Unacknowledged backlog, watermark/full-window events, oldest-unacknowledged lag, and RTT.

Receiver stress

Uncommitted samples, rejection reasons, out-of-range arrivals, and lost-by-writer events.

> Raw counts cannot compare a 1 Hz control route with a high-rate payload route. Report interval rates and ratios against offered load.

From protocol evidence to link state



Degraded

Protocol pressure persists above calibrated thresholds while the peer remains reachable: repair rate, RTT, backlog, or receive-window stress.

Unavailable

Presence/liveliness declares the peer or path dead. This is distinct from a degraded but still communicating link.

Do not mistake symptoms for root cause

Likely network symptoms

- NACK/repair growth in both directions
- increasing RTT under offered load
- inactive readers or retransmission bursts
- fragment repair on constrained MTUs

Possible local causes

- reader rejection or receive-window exhaustion
- writer cache full or history overwrite
- endpoint restart, rematch, or durable replay
- slow application draining or host resource pressure

> Protocol statistics provide evidence of reliability pressure. Packet capture, host telemetry, and controlled tests determine the cause.

Calibrate thresholds with ground truth

1. Run two Connext routers across an emulated DDIL link.
2. Sweep loss, delay, jitter, rate limits, and blackout independently.
3. Record per-peer protocol statistics, application-ack RTT, and the imposed network schedule.
4. Determine signal lag, noise floor, and threshold duration for each impairment.
5. Promote only validated rules into the degraded-link classifier.

Expected correlation

Loss drives NACKs and repair; rate limits drive backlog and window pressure; delay drives RTT and oldest-unacknowledged lag; blackout progresses from inactive readers to presence `STALE` / `DEAD`.

Link-health design summary

1. Use Connext matched-endpoint protocol status for per-peer evidence.
2. Add a dedicated application-ack probe for round-trip time.
3. Normalize counters by offered load and separate local resource symptoms.
4. Correlate measurements with controlled DDIL impairment.
5. Infer degraded links only from calibrated, sustained threshold breaches.

The router becomes the measurement point for a DDIL mesh: it relays presence, measures reliable-delivery behavior, and reports a link state that operators can act on.

References

- `docs/cpp_router/link-health.md`
- `docs/cpp_router/presence-and-health.md`
- `docs/cpp_router/design-decisions.md` (D14)
- `spikes/link_probe/README.md`
- Connext 7.7 Modern C++ protocol-status APIs

Connext AI was used for additional API context. API claims in this deck were cross-checked against the repository's header-verified link-health design and `docs/connext-ai-issues/`.